

PORT HEIDEN, AK, USA

WMO: 703330

Lat: 56.959N

Lon: 158.632W

Elev: 88

StdP: 14.65

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25508

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.9	0.2	-22.5	1.6	12.9	-15.7	2.3	18.2	43.0	38.0	39.2	38.7	15.1	50	0.870

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.6	64.8	56.8	62.7	55.1	60.5	54.0	58.7	63.8	56.5	61.0	54.8	58.9	15.3	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
55.7	66.3	60.6	54.3	62.8	59.0	52.4	58.6	57.0	25.5	64.2	24.1	61.0	23.1	59.0	65.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
36.3	31.8	27.4	-10.8	71.2	6.5	3.7	-15.5	73.8	-19.2	76.0	-22.9	78.0	-27.6	80.7
			-10.6	62.1	6.3	2.7	-15.2	64.0	-18.9	65.5	-22.5	67.0	-27.1	69.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.8	24.5	27.6	26.0	33.2	40.9	47.2	51.7	53.1	48.5	39.8	32.9	27.9
	DBStd	12.98	12.84	11.56	10.38	6.66	5.68	4.96	4.20	4.04	4.51	6.93	9.00	11.08
	HDD50	4704	792	628	743	505	287	106	30	13	78	321	514	687
	HDD65	9923	1257	1048	1208	955	748	535	413	368	495	783	963	1150
	CDD50	256	0	0	0	0	3	21	82	109	33	4	1	3
	CDD65	1	0	0	0	0	0	0	0	0	0	0	0	1
	CDH74	12	0	0	0	0	0	0	0	0	0	0	0	12
	CDH80	6	0	0	0	0	0	0	0	0	0	0	0	6
Wind	WSAvg	12.9	13.6	14.6	13.6	13.0	12.2	11.6	10.8	11.5	12.8	13.2	13.8	14.6
Precipitation	PrecAvg	35.8	3.2	2.7	2.2	1.9	2.0	2.4	3.1	4.1	3.8	3.5	3.3	3.5
	PrecMax	47.0	10.9	6.3	4.6	3.6	3.9	3.8	5.3	6.1	5.9	6.3	5.6	7.0
	PrecMin	27.5	0.8	0.5	0.4	0.5	0.4	1.3	1.2	2.2	1.9	1.3	1.1	1.3
	PrecStd	5.1	1.7	1.2	1.1	0.7	0.9	0.7	0.9	0.9	0.9	1.2	1.2	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.5	46.4	46.1	54.0	63.1	64.4	70.4	70.5	64.4	56.7	53.8	52.3
		MCWB	40.1	41.3	40.2	46.9	47.1	54.4	59.6	60.5	57.2	50.6	48.6	41.3
	2%	DB	43.3	44.6	43.3	47.6	55.3	61.3	63.9	64.8	60.7	54.2	49.9	45.7
		MCWB	38.2	39.9	37.9	41.9	46.3	51.4	56.3	58.1	54.4	49.6	46.2	40.6
	5%	DB	41.0	42.6	40.7	45.2	52.8	57.4	61.3	62.8	57.1	52.0	46.4	43.2
		MCWB	37.0	38.5	36.3	39.5	44.9	49.7	54.5	56.5	51.8	48.1	42.4	38.6
	10%	DB	39.0	39.4	38.6	43.0	49.7	54.9	58.7	60.6	55.0	49.4	44.7	40.7
		MCWB	35.5	36.2	34.7	37.7	43.7	48.6	53.2	55.2	50.6	46.0	40.6	36.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.2	42.5	41.2	47.1	50.7	55.9	60.4	61.9	59.1	51.8	49.6	45.2
		MCDB	44.6	46.0	45.1	52.9	58.3	62.6	67.7	67.9	64.0	55.2	52.7	50.8
	2%	WB	39.6	40.6	38.9	42.2	47.4	53.2	57.5	59.5	55.1	50.1	46.2	41.4
		MCDB	43.2	44.0	42.7	46.8	54.9	59.5	62.9	64.3	59.0	53.5	49.2	45.3
	5%	WB	37.8	38.5	37.0	40.1	45.7	51.1	55.6	57.4	52.9	48.2	43.3	39.7
		MCDB	40.5	41.9	40.5	44.5	51.6	56.6	60.2	61.4	56.3	51.5	46.1	43.2
	10%	WB	36.2	36.8	35.1	38.3	44.0	49.5	54.1	55.7	51.2	46.2	40.9	37.5
		MCDB	38.8	39.9	38.3	42.5	49.0	54.5	57.5	59.6	54.5	48.9	44.2	40.5
Mean Daily Temperature Range	5% DB	MDBR	10.6	10.9	11.3	11.2	11.8	11.5	9.9	9.6	10.3	10.2	9.9	10.6
		MCDBR	9.3	10.8	12.1	14.0	17.7	17.7	15.3	15.0	13.9	11.7	10.3	10.3
		MCWBR	7.7	8.5	9.3	10.3	11.3	11.0	9.8	9.5	9.5	8.8	8.6	8.3
	5% WB	MCDBR	8.8	10.4	11.9	13.4	15.4	15.6	13.8	13.7	12.0	10.9	10.7	10.3
		MCWBR	7.6	8.5	9.4	10.0	10.2	10.1	9.2	9.2	9.0	8.8	9.3	8.4
Clear-Sky Solar Irradiance	taub	0.290	0.334	0.354	0.389	0.401	0.395	0.379	0.379	0.358	0.331	0.318	0.279	
	taud	2.193	2.192	2.133	2.142	2.203	2.314	2.425	2.415	2.418	2.349	2.137	2.192	
	Ebn at Noon	182	221	249	259	265	268	271	263	251	223	162	153	
	Edh at Noon	17	26	36	41	41	38	33	31	27	22	18	13	
All-Sky Solar Radiation	RadAvg	134	303	742	1147	1446	1551	1344	1112	781	419	171	91	
	RadStd	18	39	111	105	145	130	131	114	75	32	19	7	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page